

Lab 1 – Requirements Engineering & UML Use-Case Modelling

Project Title: Hostel Maintenance & Issue Ticketing System

Domain: Campus & Academic Operations

Actors: Hostel Resident, Maintenance Warden

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SRN: PES1UG24CS106

Subject: Software Engineering

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a hostel resident to report a maintenance issue by entering the room number, issue category, and issue description.	High	When all required details are entered and the resident selects “Submit,” the system creates a ticket and displays a unique ticket ID.	Core functionality for reporting hostel issues.
FR-002	Functional	The system shall allow a maintenance warden to view submitted maintenance issue tickets.	High	When the warden opens the ticket list, all successfully submitted tickets are displayed with their ticket ID, room number, category, and status.	Enables the warden to monitor reported issues.
FR-003	Functional	The system shall allow the maintenance warden to assign a submitted maintenance issue ticket to a maintenance staff member.	High	When the warden assigns a ticket, the assigned maintenance staff member is recorded and the ticket status changes to Assigned .	Ensures reported issues are assigned to the appropriate maintenance personnel.

FR-004	Functional	The system shall allow maintenance staff to update the status of an assigned maintenance issue ticket.	High	When maintenance staff update a ticket, its status can be changed to In Progress or Resolved , and the new status is saved.	Allows maintenance work to be tracked from assignment through resolution.
FR-005	Functional	The system shall allow a hostel resident to view the current status of their submitted maintenance issue tickets.	Medium	When a resident selects a submitted ticket, the system displays its ticket ID and current status.	Keeps residents informed about the progress of their reported issues.
NFR-001	Nonfunctional	The system shall display the maintenance ticket submission confirmation within 3 seconds of a valid submission.	High	The confirmation message is displayed within 3 seconds for at least 95 out of 100 valid ticket submissions.	Ensures quick feedback to residents.
NFR-002	Nonfunctional	The system shall restrict access to maintenance ticket information based on the user's role.	High	A hostel resident cannot access another resident's ticket information, while the maintenance warden can access all submitted tickets.	Protects ticket information from unauthorized access.